

for $r \neq 0$. If we hold θ constant and let $r \rightarrow 0$, the limit is 0. On the path $y = x^2$, however, we have $r \sin \theta = r^2 \cos^2 \theta$ and

$$\begin{aligned} f(r \cos \theta, r \sin \theta) &= \frac{r \cos \theta \sin 2\theta}{r^2 \cos^4 \theta + (r \cos^2 \theta)^2} \\ &= \frac{2r \cos^2 \theta \sin \theta}{2r^2 \cos^4 \theta} = \frac{r \sin \theta}{r^2 \cos^2 \theta} = 1. \end{aligned}$$

In Exercises 61–66, find the limit of f as $(x, y) \rightarrow (0, 0)$ or show that the limit does not exist.

61. $f(x, y) = \frac{x^3 - xy^2}{x^2 + y^2}$ 62. $f(x, y) = \cos\left(\frac{x^3 - y^3}{x^2 + y^2}\right)$
 63. $f(x, y) = \frac{y^2}{x^2 + y^2}$ 64. $f(x, y) = \frac{2x}{x^2 + x + y^2}$
 65. $f(x, y) = \tan^{-1}\left(\frac{|x| + |y|}{x^2 + y^2}\right)$
 66. $f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}$

In Exercises 67 and 68, define $f(0, 0)$ in a way that extends f to be continuous at the origin.

67. $f(x, y) = \ln\left(\frac{3x^2 - x^2y^2 + 3y^2}{x^2 + y^2}\right)$
 68. $f(x, y) = \frac{3x^2y}{x^2 + y^2}$

Using the Limit Definition

Each of Exercises 69–74 gives a function $f(x, y)$ and a positive number ϵ . In each exercise, show that there exists a $\delta > 0$ such that for all (x, y) ,

$$\sqrt{x^2 + y^2} < \delta \implies |f(x, y) - f(0, 0)| < \epsilon.$$

69. $f(x, y) = x^2 + y^2$, $\epsilon = 0.01$
 70. $f(x, y) = y/(x^2 + 1)$, $\epsilon = 0.05$
 71. $f(x, y) = (x + y)/(x^2 + 1)$, $\epsilon = 0.01$
 72. $f(x, y) = (x + y)/(2 + \cos x)$, $\epsilon = 0.02$
 73. $f(x, y) = \frac{xy^2}{x^2 + y^2}$ and $f(0, 0) = 0$, $\epsilon = 0.04$
 74. $f(x, y) = \frac{x^3 + y^4}{x^2 + y^2}$ and $f(0, 0) = 0$, $\epsilon = 0.02$

Each of Exercises 75–78 gives a function $f(x, y, z)$ and a positive number ϵ . In each exercise, show that there exists a $\delta > 0$ such that for all (x, y, z) ,

$$\sqrt{x^2 + y^2 + z^2} < \delta \implies |f(x, y, z) - f(0, 0, 0)| < \epsilon.$$

75. $f(x, y, z) = x^2 + y^2 + z^2$, $\epsilon = 0.015$
 76. $f(x, y, z) = xyz$, $\epsilon = 0.008$
 77. $f(x, y, z) = \frac{x + y + z}{x^2 + y^2 + z^2 + 1}$, $\epsilon = 0.015$
 78. $f(x, y, z) = \tan^2 x + \tan^2 y + \tan^2 z$, $\epsilon = 0.03$
 79. Show that $f(x, y, z) = x + y - z$ is continuous at every point (x_0, y_0, z_0) .
 80. Show that $f(x, y, z) = x^2 + y^2 + z^2$ is continuous at the origin.

14.3 Partial Derivatives

The calculus of several variables is similar to single-variable calculus applied to several variables one at a time. When we hold all but one of the independent variables of a function constant and differentiate with respect to that one variable, we get a “partial” derivative. This section shows how partial derivatives are defined and interpreted geometrically, and how to calculate them by applying the rules for differentiating functions of a single variable. The idea of *differentiability* for functions of several variables requires more than the existence of the partial derivatives, but we will see that differentiable functions of several variables behave in the same way as differentiable single-variable functions.

Partial Derivatives of a Function of Two Variables

If (x_0, y_0) is a point in the domain of a function $f(x, y)$, the vertical plane $y = y_0$ will cut the surface $z = f(x, y)$ in the curve $z = f(x, y_0)$ (Figure 14.15). This curve is the graph of the function $z = f(x, y_0)$ in the plane $y = y_0$. The horizontal coordinate in this plane is x ; the vertical coordinate is z . The y -value is held constant at y_0 , so y is not a variable.

We define the partial derivative of f with respect to x at the point (x_0, y_0) as the ordinary derivative of $f(x, y_0)$ with respect to x at the point $x = x_0$. To distinguish partial derivatives from ordinary derivatives we use the symbol ∂ rather than the d previously used. In the definition, h represents a real number, positive or negative.

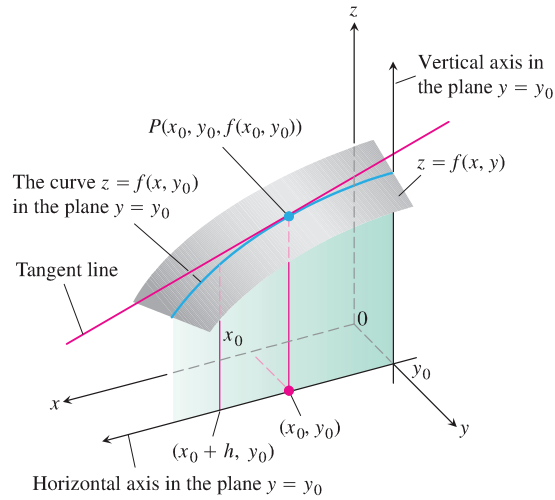


FIGURE 14.15 The intersection of the plane $y = y_0$ with the surface $z = f(x, y)$, viewed from above the first quadrant of the xy -plane.

DEFINITION The **partial derivative of $f(x, y)$ with respect to x** at the point (x_0, y_0) is

$$\left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)} = \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h},$$

provided the limit exists.

An equivalent expression for the partial derivative is

$$\left. \frac{d}{dx} f(x, y_0) \right|_{x=x_0}.$$

The slope of the curve $z = f(x, y_0)$ at the point $P(x_0, y_0, f(x_0, y_0))$ in the plane $y = y_0$ is the value of the partial derivative of f with respect to x at (x_0, y_0) . (In Figure 14.15 this slope is negative.) The tangent line to the curve at P is the line in the plane $y = y_0$ that passes through P with this slope. The partial derivative $\partial f / \partial x$ at (x_0, y_0) gives the rate of change of f with respect to x when y is held fixed at the value y_0 .

We use several notations for the partial derivative:

$$\left. \frac{\partial f}{\partial x} \right|_{(x_0, y_0)} \text{ or } f_x(x_0, y_0), \quad \left. \frac{\partial z}{\partial x} \right|_{(x_0, y_0)}, \quad \text{and} \quad f_x, \quad \frac{\partial f}{\partial x}, \quad z_x, \quad \text{or} \quad \frac{\partial z}{\partial x}.$$

The definition of the partial derivative of $f(x, y)$ with respect to y at a point (x_0, y_0) is similar to the definition of the partial derivative of f with respect to x . We hold x fixed at the value x_0 and take the ordinary derivative of $f(x_0, y)$ with respect to y at y_0 .

DEFINITION The **partial derivative of $f(x, y)$ with respect to y** at the point (x_0, y_0) is

$$\left. \frac{\partial f}{\partial y} \right|_{(x_0, y_0)} = \left. \frac{d}{dy} f(x_0, y) \right|_{y=y_0} = \lim_{h \rightarrow 0} \frac{f(x_0, y_0 + h) - f(x_0, y_0)}{h},$$

provided the limit exists.

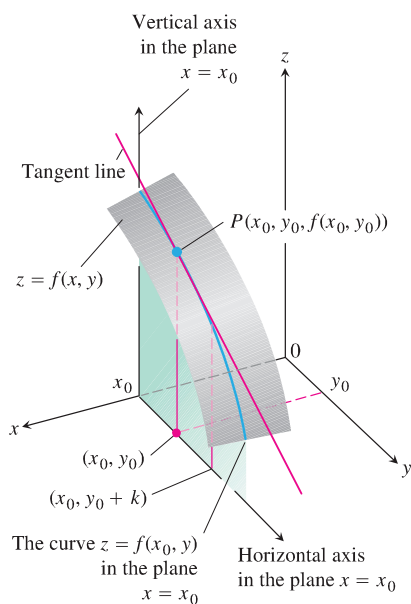


FIGURE 14.16 The intersection of the plane $x = x_0$ with the surface $z = f(x, y)$, viewed from above the first quadrant of the xy -plane.

The slope of the curve $z = f(x_0, y)$ at the point $P(x_0, y_0, f(x_0, y_0))$ in the vertical plane $x = x_0$ (Figure 14.16) is the partial derivative of f with respect to y at (x_0, y_0) . The tangent line to the curve at P is the line in the plane $x = x_0$ that passes through P with this slope. The partial derivative gives the rate of change of f with respect to y at (x_0, y_0) when x is held fixed at the value x_0 .

The partial derivative with respect to y is denoted the same way as the partial derivative with respect to x :

$$\frac{\partial f}{\partial y}(x_0, y_0), \quad f_y(x_0, y_0), \quad \frac{\partial f}{\partial y}, \quad f_y.$$

Notice that we now have two tangent lines associated with the surface $z = f(x, y)$ at the point $P(x_0, y_0, f(x_0, y_0))$ (Figure 14.17). Is the plane they determine tangent to the surface at P ? We will see that it is for the *differentiable* functions defined at the end of this section, and we will learn how to find the tangent plane in Section 14.6. First we have to learn more about partial derivatives themselves.

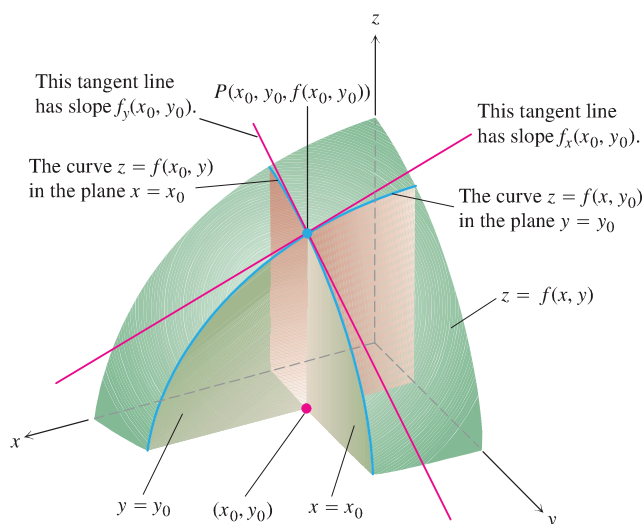


FIGURE 14.17 Figures 14.15 and 14.16 combined. The tangent lines at the point $(x_0, y_0, f(x_0, y_0))$ determine a plane that, in this picture at least, appears to be tangent to the surface.

Calculations

The definitions of $\partial f/\partial x$ and $\partial f/\partial y$ give us two different ways of differentiating f at a point: with respect to x in the usual way while treating y as a constant and with respect to y in the usual way while treating x as a constant. As the following examples show, the values of these partial derivatives are usually different at a given point (x_0, y_0) .

EXAMPLE 1 Find the values of $\partial f/\partial x$ and $\partial f/\partial y$ at the point $(4, -5)$ if

$$f(x, y) = x^2 + 3xy + y - 1.$$

Solution To find $\partial f/\partial x$, we treat y as a constant and differentiate with respect to x :

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(x^2 + 3xy + y - 1) = 2x + 3 \cdot 1 \cdot y + 0 - 0 = 2x + 3y.$$

The value of $\partial f/\partial x$ at $(4, -5)$ is $2(4) + 3(-5) = -7$.

To find $\partial f/\partial y$, we treat x as a constant and differentiate with respect to y :

$$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(x^2 + 3xy + y - 1) = 0 + 3 \cdot x \cdot 1 + 1 - 0 = 3x + 1.$$

The value of $\partial f/\partial y$ at $(4, -5)$ is $3(4) + 1 = 13$. ■

EXAMPLE 2 Find $\partial f/\partial y$ as a function if $f(x, y) = y \sin xy$.

Solution We treat x as a constant and f as a product of y and $\sin xy$:

$$\begin{aligned}\frac{\partial f}{\partial y} &= \frac{\partial}{\partial y}(y \sin xy) = y \frac{\partial}{\partial y} \sin xy + (\sin xy) \frac{\partial}{\partial y}(y) \\ &= (y \cos xy) \frac{\partial}{\partial y}(xy) + \sin xy = xy \cos xy + \sin xy.\end{aligned}$$
 ■

EXAMPLE 3 Find f_x and f_y as functions if

$$f(x, y) = \frac{2y}{y + \cos x}.$$

Solution We treat f as a quotient. With y held constant, we get

$$\begin{aligned}f_x &= \frac{\partial}{\partial x} \left(\frac{2y}{y + \cos x} \right) = \frac{(y + \cos x) \frac{\partial}{\partial x}(2y) - 2y \frac{\partial}{\partial x}(y + \cos x)}{(y + \cos x)^2} \\ &= \frac{(y + \cos x)(0) - 2y(-\sin x)}{(y + \cos x)^2} = \frac{2y \sin x}{(y + \cos x)^2}.\end{aligned}$$

With x held constant, we get

$$\begin{aligned}f_y &= \frac{\partial}{\partial y} \left(\frac{2y}{y + \cos x} \right) = \frac{(y + \cos x) \frac{\partial}{\partial y}(2y) - 2y \frac{\partial}{\partial y}(y + \cos x)}{(y + \cos x)^2} \\ &= \frac{(y + \cos x)(2) - 2y(1)}{(y + \cos x)^2} = \frac{2 \cos x}{(y + \cos x)^2}.\end{aligned}$$
 ■

Implicit differentiation works for partial derivatives the way it works for ordinary derivatives, as the next example illustrates.

EXAMPLE 4 Find $\partial z/\partial x$ if the equation

$$yz - \ln z = x + y$$

defines z as a function of the two independent variables x and y and the partial derivative exists.

Solution We differentiate both sides of the equation with respect to x , holding y constant and treating z as a differentiable function of x :

$$\begin{aligned}\frac{\partial}{\partial x}(yz) - \frac{\partial}{\partial x} \ln z &= \frac{\partial x}{\partial x} + \frac{\partial y}{\partial x} \\ y \frac{\partial z}{\partial x} - \frac{1}{z} \frac{\partial z}{\partial x} &= 1 + 0 && \text{With } y \text{ constant,} \\ &&& \frac{\partial}{\partial x}(yz) = y \frac{\partial z}{\partial x}.\end{aligned}$$

$$\begin{aligned}\left(y - \frac{1}{z}\right) \frac{\partial z}{\partial x} &= 1 \\ \frac{\partial z}{\partial x} &= \frac{z}{yz - 1}.\end{aligned}$$
 ■

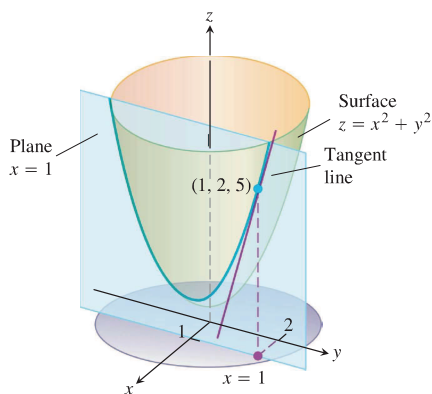


FIGURE 14.18 The tangent to the curve of intersection of the plane $x = 1$ and surface $z = x^2 + y^2$ at the point $(1, 2, 5)$ (Example 5).

EXAMPLE 5 The plane $x = 1$ intersects the paraboloid $z = x^2 + y^2$ in a parabola. Find the slope of the tangent to the parabola at $(1, 2, 5)$ (Figure 14.18).

Solution The slope is the value of the partial derivative $\partial z / \partial y$ at $(1, 2)$:

$$\left. \frac{\partial z}{\partial y} \right|_{(1,2)} = \left. \frac{\partial}{\partial y} (x^2 + y^2) \right|_{(1,2)} = 2y \Big|_{(1,2)} = 2(2) = 4.$$

As a check, we can treat the parabola as the graph of the single-variable function $z = (1)^2 + y^2 = 1 + y^2$ in the plane $x = 1$ and ask for the slope at $y = 2$. The slope, calculated now as an ordinary derivative, is

$$\left. \frac{dz}{dy} \right|_{y=2} = \left. \frac{d}{dy} (1 + y^2) \right|_{y=2} = 2y \Big|_{y=2} = 4. \quad \blacksquare$$

Functions of More Than Two Variables

The definitions of the partial derivatives of functions of more than two independent variables are like the definitions for functions of two variables. They are ordinary derivatives with respect to one variable, taken while the other independent variables are held constant.

EXAMPLE 6 If x , y , and z are independent variables and

$$f(x, y, z) = x \sin(y + 3z),$$

then

$$\begin{aligned} \frac{\partial f}{\partial z} &= \frac{\partial}{\partial z} [x \sin(y + 3z)] = x \frac{\partial}{\partial z} \sin(y + 3z) \\ &= x \cos(y + 3z) \frac{\partial}{\partial z} (y + 3z) = 3x \cos(y + 3z). \end{aligned} \quad \blacksquare$$

EXAMPLE 7 If resistors of R_1 , R_2 , and R_3 ohms are connected in parallel to make an R -ohm resistor, the value of R can be found from the equation

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

(Figure 14.19). Find the value of $\partial R / \partial R_2$ when $R_1 = 30$, $R_2 = 45$, and $R_3 = 90$ ohms.

Solution To find $\partial R / \partial R_2$, we treat R_1 and R_3 as constants and, using implicit differentiation, differentiate both sides of the equation with respect to R_2 :

$$\begin{aligned} \frac{\partial}{\partial R_2} \left(\frac{1}{R} \right) &= \frac{\partial}{\partial R_2} \left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \right) \\ -\frac{1}{R^2} \frac{\partial R}{\partial R_2} &= 0 - \frac{1}{R_2^2} + 0 \\ \frac{\partial R}{\partial R_2} &= \frac{R^2}{R_2^2} = \left(\frac{R}{R_2} \right)^2. \end{aligned}$$

When $R_1 = 30$, $R_2 = 45$, and $R_3 = 90$,

$$\frac{1}{R} = \frac{1}{30} + \frac{1}{45} + \frac{1}{90} = \frac{3 + 2 + 1}{90} = \frac{6}{90} = \frac{1}{15},$$

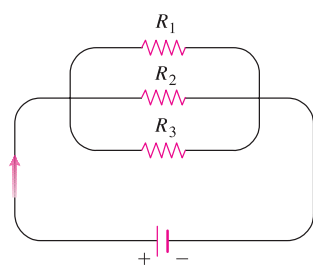


FIGURE 14.19 Resistors arranged this way are said to be connected in parallel (Example 7). Each resistor lets a portion of the current through. Their equivalent resistance R is calculated with the formula

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}.$$

so $R = 15$ and

$$\frac{\partial R}{\partial R_2} = \left(\frac{15}{45}\right)^2 = \left(\frac{1}{3}\right)^2 = \frac{1}{9}.$$

Thus at the given values, a small change in the resistance R_2 leads to a change in R about $1/9$ th as large. ■

Partial Derivatives and Continuity

A function $f(x, y)$ can have partial derivatives with respect to both x and y at a point without the function being continuous there. This is different from functions of a single variable, where the existence of a derivative implies continuity. If the partial derivatives of $f(x, y)$ exist and are continuous throughout a disk centered at (x_0, y_0) , however, then f is continuous at (x_0, y_0) , as we see at the end of this section.

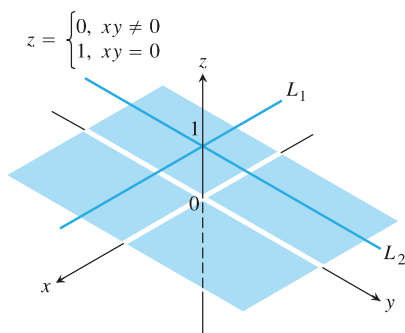


FIGURE 14.20 The graph of

$$f(x, y) = \begin{cases} 0, & xy \neq 0 \\ 1, & xy = 0 \end{cases}$$

consists of the lines L_1 and L_2 and the four open quadrants of the xy -plane. The function has partial derivatives at the origin but is not continuous there (Example 8).

EXAMPLE 8 Let

$$f(x, y) = \begin{cases} 0, & xy \neq 0 \\ 1, & xy = 0 \end{cases}$$

(Figure 14.20).

- (a) Find the limit of f as (x, y) approaches $(0, 0)$ along the line $y = x$.
- (b) Prove that f is not continuous at the origin.
- (c) Show that both partial derivatives $\partial f/\partial x$ and $\partial f/\partial y$ exist at the origin.

Solution

- (a) Since $f(x, y)$ is constantly zero along the line $y = x$ (except at the origin), we have

$$\lim_{(x, y) \rightarrow (0, 0)} f(x, y) \Big|_{y=x} = \lim_{(x, y) \rightarrow (0, 0)} 0 = 0.$$

- (b) Since $f(0, 0) = 1$, the limit in part (a) proves that f is not continuous at $(0, 0)$.
- (c) To find $\partial f/\partial x$ at $(0, 0)$, we hold y fixed at $y = 0$. Then $f(x, y) = 1$ for all x , and the graph of f is the line L_1 in Figure 14.20. The slope of this line at any x is $\partial f/\partial x = 0$. In particular, $\partial f/\partial x = 0$ at $(0, 0)$. Similarly, $\partial f/\partial y$ is the slope of line L_2 at any y , so $\partial f/\partial y = 0$ at $(0, 0)$. ■

Example 8 notwithstanding, it is still true in higher dimensions that *differentiability* at a point implies continuity. What Example 8 suggests is that we need a stronger requirement for differentiability in higher dimensions than the mere existence of the partial derivatives. We define differentiability for functions of two variables (which is slightly more complicated than for single-variable functions) at the end of this section and then revisit the connection to continuity.

Second-Order Partial Derivatives

When we differentiate a function $f(x, y)$ twice, we produce its second-order derivatives. These derivatives are usually denoted by

$$\begin{aligned} \frac{\partial^2 f}{\partial x^2} \text{ or } f_{xx}, & \quad \frac{\partial^2 f}{\partial y^2} \text{ or } f_{yy}, \\ \frac{\partial^2 f}{\partial x \partial y} \text{ or } f_{yx}, & \quad \text{and} \quad \frac{\partial^2 f}{\partial y \partial x} \text{ or } f_{xy}. \end{aligned}$$

The defining equations are

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right), \quad \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right),$$

and so on. Notice the order in which the mixed partial derivatives are taken:

$$\frac{\partial^2 f}{\partial x \partial y} \quad \text{Differentiate first with respect to } y, \text{ then with respect to } x.$$

$$f_{yx} = (f_y)_x \quad \text{Means the same thing.}$$

HISTORICAL BIOGRAPHY

Pierre-Simon Laplace
(1749–1827)

EXAMPLE 9 If $f(x, y) = x \cos y + ye^x$, find the second-order derivatives

$$\frac{\partial^2 f}{\partial x^2}, \quad \frac{\partial^2 f}{\partial y \partial x}, \quad \frac{\partial^2 f}{\partial y^2}, \quad \text{and} \quad \frac{\partial^2 f}{\partial x \partial y}.$$

Solution The first step is to calculate both first partial derivatives.

$$\begin{aligned} \frac{\partial f}{\partial x} &= \frac{\partial}{\partial x} (x \cos y + ye^x) & \frac{\partial f}{\partial y} &= \frac{\partial}{\partial y} (x \cos y + ye^x) \\ &= \cos y + ye^x & &= -x \sin y + e^x \end{aligned}$$

Now we find both partial derivatives of each first partial:

$$\begin{aligned} \frac{\partial^2 f}{\partial y \partial x} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = -\sin y + e^x & \frac{\partial^2 f}{\partial x \partial y} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = -\sin y + e^x \\ \frac{\partial^2 f}{\partial x^2} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = ye^x & \frac{\partial^2 f}{\partial y^2} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = -x \cos y. \end{aligned}$$

The Mixed Derivative Theorem

You may have noticed that the “mixed” second-order partial derivatives

$$\frac{\partial^2 f}{\partial y \partial x} \quad \text{and} \quad \frac{\partial^2 f}{\partial x \partial y}$$

in Example 9 are equal. This is not a coincidence. They must be equal whenever f , f_x , f_y , f_{xy} , and f_{yx} are continuous, as stated in the following theorem.

THEOREM 2—The Mixed Derivative Theorem If $f(x, y)$ and its partial derivatives f_x , f_y , f_{xy} , and f_{yx} are defined throughout an open region containing a point (a, b) and are all continuous at (a, b) , then

$$f_{xy}(a, b) = f_{yx}(a, b).$$

HISTORICAL BIOGRAPHY

Alexis Clairaut
(1713–1765)

Theorem 2 is also known as Clairaut’s Theorem, named after the French mathematician Alexis Clairaut who discovered it. A proof is given in Appendix 9. Theorem 2 says that to calculate a mixed second-order derivative, we may differentiate in either order, provided the continuity conditions are satisfied. This ability to proceed in different order sometimes simplifies our calculations.

EXAMPLE 10 Find $\partial^2 w / \partial x \partial y$ if

$$w = xy + \frac{e^y}{y^2 + 1}.$$

Solution The symbol $\partial^2 w / \partial x \partial y$ tells us to differentiate first with respect to y and then with respect to x . However, if we interchange the order of differentiation and differentiate first with respect to x we get the answer more quickly. In two steps,

$$\frac{\partial w}{\partial x} = y \quad \text{and} \quad \frac{\partial^2 w}{\partial y \partial x} = 1.$$

If we differentiate first with respect to y , we obtain $\partial^2 w / \partial x \partial y = 1$ as well. We can differentiate in either order because the conditions of Theorem 2 hold for w at all points (x_0, y_0) . ■

Partial Derivatives of Still Higher Order

Although we will deal mostly with first- and second-order partial derivatives, because these appear the most frequently in applications, there is no theoretical limit to how many times we can differentiate a function as long as the derivatives involved exist. Thus, we get third- and fourth-order derivatives denoted by symbols like

$$\frac{\partial^3 f}{\partial x \partial y^2} = f_{yyx},$$

$$\frac{\partial^4 f}{\partial x^2 \partial y^2} = f_{yyxx},$$

and so on. As with second-order derivatives, the order of differentiation is immaterial as long as all the derivatives through the order in question are continuous.

EXAMPLE 11 Find f_{xyz} if $f(x, y, z) = 1 - 2xy^2z + x^2y$.

Solution We first differentiate with respect to the variable y , then x , then y again, and finally with respect to z :

$$f_y = -4xyz + x^2$$

$$f_{yx} = -4yz + 2x$$

$$f_{yxy} = -4z$$

$$f_{yxzy} = -4$$

■

Differentiability

The starting point for differentiability is not the difference quotient we saw in studying single-variable functions, but rather the idea of increment. Recall from our work with functions of a single variable in Section 3.9 that if $y = f(x)$ is differentiable at $x = x_0$, then the change in the value of f that results from changing x from x_0 to $x_0 + \Delta x$ is given by an equation of the form

$$\Delta y = f'(x_0)\Delta x + \epsilon \Delta x$$

in which $\epsilon \rightarrow 0$ as $\Delta x \rightarrow 0$. For functions of two variables, the analogous property becomes the definition of differentiability. The Increment Theorem (proved in Appendix 9) tells us when to expect the property to hold.

THEOREM 3—The Increment Theorem for Functions of Two Variables Suppose that the first partial derivatives of $f(x, y)$ are defined throughout an open region R containing the point (x_0, y_0) and that f_x and f_y are continuous at (x_0, y_0) . Then the change

$$\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$$

in the value of f that results from moving from (x_0, y_0) to another point $(x_0 + \Delta x, y_0 + \Delta y)$ in R satisfies an equation of the form

$$\Delta z = f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y + \epsilon_1\Delta x + \epsilon_2\Delta y$$

in which each of $\epsilon_1, \epsilon_2 \rightarrow 0$ as both $\Delta x, \Delta y \rightarrow 0$.

You can see where the epsilons come from in the proof given in Appendix 9. Similar results hold for functions of more than two independent variables.

DEFINITION A function $z = f(x, y)$ is **differentiable at** (x_0, y_0) if $f_x(x_0, y_0)$ and $f_y(x_0, y_0)$ exist and Δz satisfies an equation of the form

$$\Delta z = f_x(x_0, y_0)\Delta x + f_y(x_0, y_0)\Delta y + \epsilon_1\Delta x + \epsilon_2\Delta y$$

in which each of $\epsilon_1, \epsilon_2 \rightarrow 0$ as both $\Delta x, \Delta y \rightarrow 0$. We call f **differentiable** if it is differentiable at every point in its domain, and say that its graph is a **smooth surface**.

Because of this definition, an immediate corollary of Theorem 3 is that a function is differentiable at (x_0, y_0) if its first partial derivatives are *continuous* there.

COROLLARY OF THEOREM 3 If the partial derivatives f_x and f_y of a function $f(x, y)$ are continuous throughout an open region R , then f is differentiable at every point of R .

If $z = f(x, y)$ is differentiable, then the definition of differentiability assures that $\Delta z = f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)$ approaches 0 as Δx and Δy approach 0. This tells us that a function of two variables is continuous at every point where it is differentiable.

THEOREM 4—Differentiability Implies Continuity If a function $f(x, y)$ is differentiable at (x_0, y_0) , then f is continuous at (x_0, y_0) .

As we can see from Corollary 3 and Theorem 4, a function $f(x, y)$ must be continuous at a point (x_0, y_0) if f_x and f_y are continuous throughout an open region containing (x_0, y_0) . Remember, however, that it is still possible for a function of two variables to be discontinuous at a point where its first partial derivatives exist, as we saw in Example 8. Existence alone of the partial derivatives at that point is not enough, but continuity of the partial derivatives guarantees differentiability.

Exercises 14.3

Calculating First-Order Partial Derivatives

In Exercises 1–22, find $\partial f/\partial x$ and $\partial f/\partial y$.

- $f(x, y) = 2x^2 - 3y - 4$
- $f(x, y) = x^2 - xy + y^2$
- $f(x, y) = (x^2 - 1)(y + 2)$
- $f(x, y) = 5xy - 7x^2 - y^2 + 3x - 6y + 2$
- $f(x, y) = (xy - 1)^2$
- $f(x, y) = (2x - 3y)^3$
- $f(x, y) = \sqrt{x^2 + y^2}$
- $f(x, y) = (x^3 + (y/2))^{2/3}$
- $f(x, y) = 1/(x + y)$
- $f(x, y) = x/(x^2 + y^2)$
- $f(x, y) = (x + y)/(xy - 1)$
- $f(x, y) = \tan^{-1}(y/x)$
- $f(x, y) = e^{(x+y+1)}$
- $f(x, y) = e^{-x} \sin(x + y)$
- $f(x, y) = \ln(x + y)$
- $f(x, y) = e^{xy} \ln y$
- $f(x, y) = \sin^2(x - 3y)$
- $f(x, y) = \cos^2(3x - y^2)$

$$19. f(x, y) = x^y \qquad 20. f(x, y) = \log_y x$$

$$21. f(x, y) = \int_x^y g(t) dt \quad (g \text{ continuous for all } t)$$

$$22. f(x, y) = \sum_{n=0}^{\infty} (xy)^n \quad (|xy| < 1)$$

In Exercises 23–34, find f_x , f_y , and f_z .

$$23. f(x, y, z) = 1 + xy^2 - 2z^2 \quad 24. f(x, y, z) = xy + yz + xz$$

$$25. f(x, y, z) = x - \sqrt{y^2 + z^2}$$

$$26. f(x, y, z) = (x^2 + y^2 + z^2)^{-1/2}$$

$$27. f(x, y, z) = \sin^{-1}(xyz) \quad 28. f(x, y, z) = \sec^{-1}(x + yz)$$

$$29. f(x, y, z) = \ln(x + 2y + 3z)$$

30. $f(x, y, z) = yz \ln(xy)$ 31. $f(x, y, z) = e^{-(x^2+y^2+z^2)}$
 32. $f(x, y, z) = e^{-xyz}$
 33. $f(x, y, z) = \tanh(x + 2y + 3z)$
 34. $f(x, y, z) = \sinh(xy - z^2)$

In Exercises 35–40, find the partial derivative of the function with respect to each variable.

35. $f(t, \alpha) = \cos(2\pi t - \alpha)$ 36. $g(u, v) = v^2 e^{(2u/v)}$
 37. $h(\rho, \phi, \theta) = \rho \sin \phi \cos \theta$ 38. $g(r, \theta, z) = r(1 - \cos \theta) - z$
 39. **Work done by the heart** (Section 3.9, Exercise 49)

$$W(P, V, \delta, v, g) = PV + \frac{V\delta v^2}{2g}$$

40. **Wilson lot size formula** (Section 4.5, Exercise 51)

$$A(c, h, k, m, q) = \frac{km}{q} + cm + \frac{hq}{2}$$

Calculating Second-Order Partial Derivatives

Find all the second-order partial derivatives of the functions in Exercises 41–50.

41. $f(x, y) = x + y + xy$ 42. $f(x, y) = \sin xy$
 43. $g(x, y) = x^2y + \cos y + y \sin x$
 44. $h(x, y) = xe^y + y + 1$ 45. $r(x, y) = \ln(x + y)$
 46. $s(x, y) = \tan^{-1}(y/x)$ 47. $w = x^2 \tan(xy)$
 48. $w = ye^{x^2-y}$ 49. $w = x \sin(x^2y)$
 50. $w = \frac{x - y}{x^2 + y}$

Mixed Partial Derivatives

In Exercises 51–54, verify that $w_{xy} = w_{yx}$.

51. $w = \ln(2x + 3y)$ 52. $w = e^x + x \ln y + y \ln x$
 53. $w = xy^2 + x^2y^3 + x^3y^4$ 54. $w = x \sin y + y \sin x + xy$
 55. Which order of differentiation will calculate f_{xy} faster: x first or y first? Try to answer without writing anything down.
 a. $f(x, y) = x \sin y + e^y$
 b. $f(x, y) = 1/y$
 c. $f(x, y) = y + (x/y)$
 d. $f(x, y) = y + x^2y + 4y^3 - \ln(y^2 + 1)$
 e. $f(x, y) = x^2 + 5xy + \sin x + 7e^x$
 f. $f(x, y) = x \ln xy$
 56. The fifth-order partial derivative $\partial^5 f / \partial x^2 \partial y^3$ is zero for each of the following functions. To show this as quickly as possible, which variable would you differentiate with respect to first: x or y ? Try to answer without writing anything down.
 a. $f(x, y) = y^2 x^4 e^x + 2$
 b. $f(x, y) = y^2 + y(\sin x - x^4)$
 c. $f(x, y) = x^2 + 5xy + \sin x + 7e^x$
 d. $f(x, y) = xe^{y^2/2}$

Using the Partial Derivative Definition

In Exercises 57–60, use the limit definition of partial derivative to compute the partial derivatives of the functions at the specified points.

57. $f(x, y) = 1 - x + y - 3x^2y$, $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at $(1, 2)$
 58. $f(x, y) = 4 + 2x - 3y - xy^2$, $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at $(-2, 1)$
 59. $f(x, y) = \sqrt{2x + 3y - 1}$, $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at $(-2, 3)$
 60. $f(x, y) = \begin{cases} \frac{\sin(x^3 + y^4)}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$
 $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at $(0, 0)$

61. Let $f(x, y) = 2x + 3y - 4$. Find the slope of the line tangent to this surface at the point $(2, -1)$ and lying in the **a.** plane $x = 2$ **b.** plane $y = -1$.
 62. Let $f(x, y) = x^2 + y^3$. Find the slope of the line tangent to this surface at the point $(-1, 1)$ and lying in the **a.** plane $x = -1$ **b.** plane $y = 1$.
 63. **Three variables** Let $w = f(x, y, z)$ be a function of three independent variables and write the formal definition of the partial derivative $\partial f / \partial z$ at (x_0, y_0, z_0) . Use this definition to find $\partial f / \partial z$ at $(1, 2, 3)$ for $f(x, y, z) = x^2 y z^2$.
 64. **Three variables** Let $w = f(x, y, z)$ be a function of three independent variables and write the formal definition of the partial derivative $\partial f / \partial y$ at (x_0, y_0, z_0) . Use this definition to find $\partial f / \partial y$ at $(-1, 0, 3)$ for $f(x, y, z) = -2xy^2 + yz^2$.

Differentiating Implicitly

65. Find the value of $\partial z / \partial x$ at the point $(1, 1, 1)$ if the equation

$$xy + z^3x - 2yz = 0$$

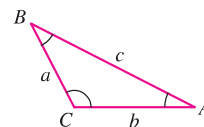
defines z as a function of the two independent variables x and y and the partial derivative exists.

66. Find the value of $\partial x / \partial z$ at the point $(1, -1, -3)$ if the equation

$$xz + y \ln x - x^2 + 4 = 0$$

defines x as a function of the two independent variables y and z and the partial derivative exists.

Exercises 67 and 68 are about the triangle shown here.



67. Express A implicitly as a function of a , b , and c and calculate $\partial A / \partial a$ and $\partial A / \partial b$.
 68. Express a implicitly as a function of A , b , and B and calculate $\partial a / \partial A$ and $\partial a / \partial B$.
 69. **Two dependent variables** Express v_x in terms of u and y if the equations $x = v \ln u$ and $y = u \ln v$ define u and v as functions of the independent variables x and y , and if v_x exists. (Hint: Differentiate both equations with respect to x and solve for v_x by eliminating u_x .)

70. Two dependent variables Find $\partial x/\partial u$ and $\partial y/\partial u$ if the equations $u = x^2 - y^2$ and $v = x^2 + y^2$ define x and y as functions of the independent variables u and v , and the partial derivatives exist. (See the hint in Exercise 69.) Then let $s = x^2 + y^2$ and find $\partial s/\partial u$.

71. Let $f(x, y) = \begin{cases} y^3, & y \geq 0 \\ -y^2, & y < 0. \end{cases}$

Find f_x , f_y , f_{xy} , and f_{yx} , and state the domain for each partial derivative.

72. Let $f(x, y) = \begin{cases} \sqrt{x}, & x \geq 0 \\ x^2, & x < 0. \end{cases}$

Find f_x , f_y , f_{xy} , and f_{yx} , and state the domain for each partial derivative.

Theory and Examples

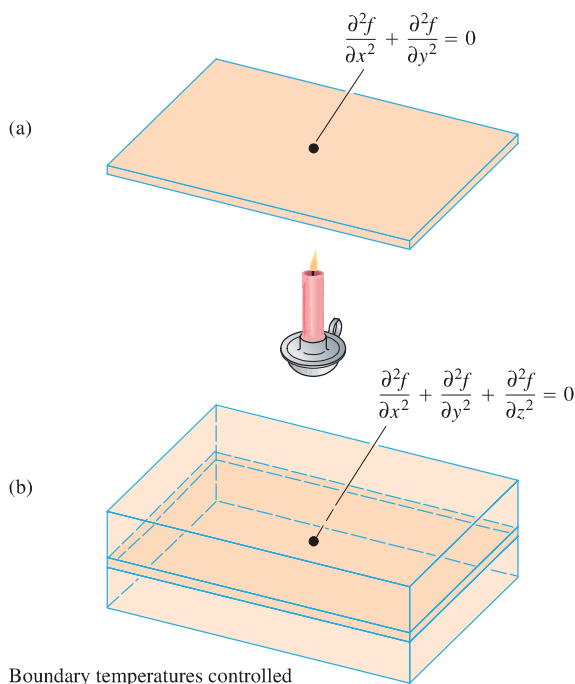
The three-dimensional Laplace equation

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$$

is satisfied by steady-state temperature distributions $T = f(x, y, z)$ in space, by gravitational potentials, and by electrostatic potentials. The **two-dimensional Laplace equation**

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0,$$

obtained by dropping the $\partial^2 f/\partial z^2$ term from the previous equation, describes potentials and steady-state temperature distributions in a plane (see the accompanying figure). The plane (a) may be treated as a thin slice of the solid (b) perpendicular to the z -axis.



Show that each function in Exercises 73–80 satisfies a Laplace equation.

73. $f(x, y, z) = x^2 + y^2 - 2z^2$

74. $f(x, y, z) = 2z^3 - 3(x^2 + y^2)z$

75. $f(x, y) = e^{-2y} \cos 2x$

76. $f(x, y) = \ln \sqrt{x^2 + y^2}$

77. $f(x, y) = 3x + 2y - 4$

78. $f(x, y) = \tan^{-1} \frac{x}{y}$

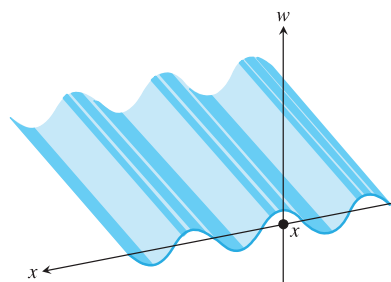
79. $f(x, y, z) = (x^2 + y^2 + z^2)^{-1/2}$

80. $f(x, y, z) = e^{3x+4y} \cos 5z$

The Wave Equation If we stand on an ocean shore and take a snapshot of the waves, the picture shows a regular pattern of peaks and valleys in an instant of time. We see periodic vertical motion in space, with respect to distance. If we stand in the water, we can feel the rise and fall of the water as the waves go by. We see periodic vertical motion in time. In physics, this beautiful symmetry is expressed by the **one-dimensional wave equation**

$$\frac{\partial^2 w}{\partial t^2} = c^2 \frac{\partial^2 w}{\partial x^2},$$

where w is the wave height, x is the distance variable, t is the time variable, and c is the velocity with which the waves are propagated.



In our example, x is the distance across the ocean's surface, but in other applications, x might be the distance along a vibrating string, distance through air (sound waves), or distance through space (light waves). The number c varies with the medium and type of wave.

Show that the functions in Exercises 81–87 are all solutions of the wave equation.

81. $w = \sin(x + ct)$

82. $w = \cos(2x + 2ct)$

83. $w = \sin(x + ct) + \cos(2x + 2ct)$

84. $w = \ln(2x + 2ct)$

85. $w = \tan(2x - 2ct)$

86. $w = 5 \cos(3x + 3ct) + e^{x+ct}$

87. $w = f(u)$, where f is a differentiable function of u , and $u = a(x + ct)$, where a is a constant

88. Does a function $f(x, y)$ with continuous first partial derivatives throughout an open region R have to be continuous on R ? Give reasons for your answer.

89. If a function $f(x, y)$ has continuous second partial derivatives throughout an open region R , must the first-order partial derivatives of f be continuous on R ? Give reasons for your answer.

- 90. The heat equation** An important partial differential equation that describes the distribution of heat in a region at time t can be represented by the *one-dimensional heat equation*

$$\frac{\partial f}{\partial t} = \frac{\partial^2 f}{\partial x^2}.$$

Show that $u(x, t) = \sin(\alpha x) \cdot e^{-\beta t}$ satisfies the heat equation for constants α and β .

- 91.** Let $f(x, y) = \begin{cases} \frac{xy^2}{x^2 + y^4}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0). \end{cases}$

Show that $f_x(0, 0)$ and $f_y(0, 0)$ exist, but f is not differentiable at $(0, 0)$. (*Hint:* Use Theorem 4 and show that f is not continuous at $(0, 0)$.)

- 92.** Let $f(x, y) = \begin{cases} 0, & x^2 < y < 2x^2 \\ 1, & \text{otherwise.} \end{cases}$

Show that $f_x(0, 0)$ and $f_y(0, 0)$ exist, but f is not differentiable at $(0, 0)$.

14.4 The Chain Rule

The Chain Rule for functions of a single variable studied in Section 3.6 says that when $w = f(x)$ is a differentiable function of x and $x = g(t)$ is a differentiable function of t , w is a differentiable function of t and dw/dt can be calculated by the formula

$$\frac{dw}{dt} = \frac{dw}{dx} \frac{dx}{dt}.$$

For functions of two or more variables the Chain Rule has several forms. The form depends on how many variables are involved, but once this is taken into account, it works like the Chain Rule in Section 3.6.

Functions of Two Variables

The Chain Rule formula for a differentiable function $w = f(x, y)$ when $x = x(t)$ and $y = y(t)$ are both differentiable functions of t is given in the following theorem.

Each of $\frac{\partial f}{\partial x}$, $\frac{\partial w}{\partial x}$, f_x indicates the partial derivative of f with respect to x .

THEOREM 5—Chain Rule for Functions of Two Independent Variables If $w = f(x, y)$ is differentiable and if $x = x(t)$, $y = y(t)$ are differentiable functions of t , then the composite $w = f(x(t), y(t))$ is a differentiable function of t and

$$\frac{dw}{dt} = f_x(x(t), y(t)) \cdot x'(t) + f_y(x(t), y(t)) \cdot y'(t),$$

or

$$\frac{dw}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}.$$

Proof The proof consists of showing that if x and y are differentiable at $t = t_0$, then w is differentiable at t_0 and

$$\left(\frac{dw}{dt} \right)_{t_0} = \left(\frac{\partial w}{\partial x} \right)_{P_0} \left(\frac{dx}{dt} \right)_{t_0} + \left(\frac{\partial w}{\partial y} \right)_{P_0} \left(\frac{dy}{dt} \right)_{t_0},$$

where $P_0 = (x(t_0), y(t_0))$. The subscripts indicate where each of the derivatives is to be evaluated.

Let Δx , Δy , and Δw be the increments that result from changing t from t_0 to $t_0 + \Delta t$. Since f is differentiable (see the definition in Section 14.3),

$$\Delta w = \left(\frac{\partial w}{\partial x} \right)_{P_0} \Delta x + \left(\frac{\partial w}{\partial y} \right)_{P_0} \Delta y + \epsilon_1 \Delta x + \epsilon_2 \Delta y,$$

where $\epsilon_1, \epsilon_2 \rightarrow 0$ as $\Delta x, \Delta y \rightarrow 0$. To find dw/dt , we divide this equation through by Δt and let Δt approach zero. The division gives

$$\frac{\Delta w}{\Delta t} = \left(\frac{\partial w}{\partial x} \right)_{P_0} \frac{\Delta x}{\Delta t} + \left(\frac{\partial w}{\partial y} \right)_{P_0} \frac{\Delta y}{\Delta t} + \epsilon_1 \frac{\Delta x}{\Delta t} + \epsilon_2 \frac{\Delta y}{\Delta t}.$$

Letting Δt approach zero gives

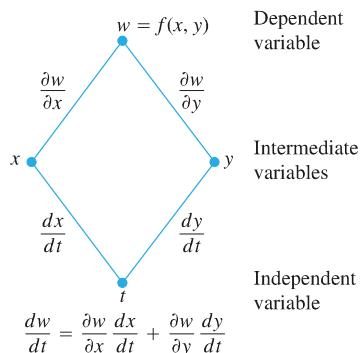
$$\begin{aligned} \left(\frac{dw}{dt} \right)_{t_0} &= \lim_{\Delta t \rightarrow 0} \frac{\Delta w}{\Delta t} \\ &= \left(\frac{\partial w}{\partial x} \right)_{P_0} \left(\frac{dx}{dt} \right)_{t_0} + \left(\frac{\partial w}{\partial y} \right)_{P_0} \left(\frac{dy}{dt} \right)_{t_0} + 0 \cdot \left(\frac{dx}{dt} \right)_{t_0} + 0 \cdot \left(\frac{dy}{dt} \right)_{t_0}. \quad \blacksquare \end{aligned}$$

Often we write $\partial w/\partial x$ for the partial derivative $\partial f/\partial x$, so we can rewrite the Chain Rule in Theorem 5 in the form

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt}.$$

To remember the Chain Rule picture the diagram below. To find dw/dt , start at w and read down each route to t , multiplying derivatives along the way. Then add the products.

Chain Rule



However, the meaning of the dependent variable w is different on each side of the preceding equation. On the left-hand side, it refers to the composite function $w = f(x(t), y(t))$ as a function of the single variable t . On the right-hand side, it refers to the function $w = f(x, y)$ as a function of the two variables x and y . Moreover, the single derivatives dw/dt , dx/dt , and dy/dt are being evaluated at a point t_0 , whereas the partial derivatives $\partial w/\partial x$ and $\partial w/\partial y$ are being evaluated at the point (x_0, y_0) , with $x_0 = x(t_0)$ and $y_0 = y(t_0)$. With that understanding, we will use both of these forms interchangeably throughout the text whenever no confusion will arise.

The **branch diagram** in the margin provides a convenient way to remember the Chain Rule. The “true” independent variable in the composite function is t , whereas x and y are *intermediate variables* (controlled by t) and w is the dependent variable.

A more precise notation for the Chain Rule shows where the various derivatives in Theorem 5 are evaluated:

$$\frac{dw}{dt}(t_0) = \frac{\partial f}{\partial x}(x_0, y_0) \cdot \frac{dx}{dt}(t_0) + \frac{\partial f}{\partial y}(x_0, y_0) \cdot \frac{dy}{dt}(t_0).$$

EXAMPLE 1 Use the Chain Rule to find the derivative of

$$w = xy$$

with respect to t along the path $x = \cos t, y = \sin t$. What is the derivative's value at $t = \pi/2$?

Solution We apply the Chain Rule to find dw/dt as follows:

$$\begin{aligned} \frac{dw}{dt} &= \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} \\ &= \frac{\partial(xy)}{\partial x} \cdot \frac{d}{dt}(\cos t) + \frac{\partial(xy)}{\partial y} \cdot \frac{d}{dt}(\sin t) \\ &= (y)(-\sin t) + (x)(\cos t) \\ &= (\sin t)(-\sin t) + (\cos t)(\cos t) \\ &= -\sin^2 t + \cos^2 t \\ &= \cos 2t. \end{aligned}$$

In this example, we can check the result with a more direct calculation. As a function of t ,

$$w = xy = \cos t \sin t = \frac{1}{2} \sin 2t,$$

so

$$\frac{dw}{dt} = \frac{d}{dt} \left(\frac{1}{2} \sin 2t \right) = \frac{1}{2} \cdot 2 \cos 2t = \cos 2t.$$

In either case, at the given value of t ,

$$\left(\frac{dw}{dt} \right)_{t=\pi/2} = \cos \left(2 \cdot \frac{\pi}{2} \right) = \cos \pi = -1.$$

Functions of Three Variables

You can probably predict the Chain Rule for functions of three variables, as it only involves adding the expected third term to the two-variable formula.

THEOREM 6—Chain Rule for Functions of Three Independent Variables If $w = f(x, y, z)$ is differentiable and x, y , and z are differentiable functions of t , then w is a differentiable function of t and

$$\frac{dw}{dt} = \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt}.$$

The proof is identical with the proof of Theorem 5 except that there are now three intermediate variables instead of two. The branch diagram we use for remembering the new equation is similar as well, with three routes from w to t .

EXAMPLE 2 Find dw/dt if

$$w = xy + z, \quad x = \cos t, \quad y = \sin t, \quad z = t.$$

In this example the values of $w(t)$ are changing along the path of a helix (Section 13.1) as t changes. What is the derivative's value at $t = 0$?

Solution Using the Chain Rule for three independent variables, we have

$$\begin{aligned} \frac{dw}{dt} &= \frac{\partial w}{\partial x} \frac{dx}{dt} + \frac{\partial w}{\partial y} \frac{dy}{dt} + \frac{\partial w}{\partial z} \frac{dz}{dt} \\ &= (y)(-\sin t) + (x)(\cos t) + (1)(1) \\ &= (\sin t)(-\sin t) + (\cos t)(\cos t) + 1 \\ &= -\sin^2 t + \cos^2 t + 1 = 1 + \cos 2t, \end{aligned}$$

Substitute for the intermediate variables.

so

$$\left(\frac{dw}{dt} \right)_{t=0} = 1 + \cos(0) = 2.$$

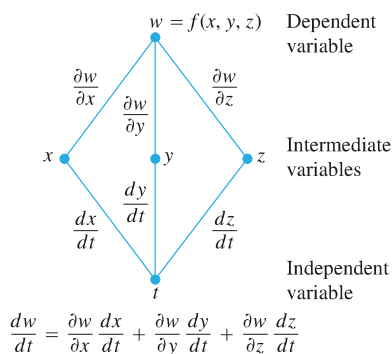
For a physical interpretation of change along a curve think of an object whose position is changing with time t . If $w = T(x, y, z)$ is the temperature at each point (x, y, z) along a curve C with parametric equations $x = x(t)$, $y = y(t)$, and $z = z(t)$, then the composite function $w = T(x(t), y(t), z(t))$ represents the temperature relative to t along the curve. The derivative dw/dt is then the instantaneous rate of change of temperature due to the motion along the curve, as calculated in Theorem 6.

Functions Defined on Surfaces

If we are interested in the temperature $w = f(x, y, z)$ at points (x, y, z) on the earth's surface, we might prefer to think of x, y , and z as functions of the variables r and s that give

Here we have three routes from w to t instead of two, but finding dw/dt is still the same. Read down each route, multiplying derivatives along the way; then add.

Chain Rule



the points' longitudes and latitudes. If $x = g(r, s)$, $y = h(r, s)$, and $z = k(r, s)$, we could then express the temperature as a function of r and s with the composite function

$$w = f(g(r, s), h(r, s), k(r, s)).$$

Under the conditions stated below, w has partial derivatives with respect to both r and s that can be calculated in the following way.

THEOREM 7—Chain Rule for Two Independent Variables and Three Intermediate Variables Suppose that $w = f(x, y, z)$, $x = g(r, s)$, $y = h(r, s)$, and $z = k(r, s)$. If all four functions are differentiable, then w has partial derivatives with respect to r and s , given by the formulas

$$\begin{aligned}\frac{\partial w}{\partial r} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial r} \\ \frac{\partial w}{\partial s} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial s} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial s}.\end{aligned}$$

The first of these equations can be derived from the Chain Rule in Theorem 6 by holding s fixed and treating r as t . The second can be derived in the same way, holding r fixed and treating s as t . The branch diagrams for both equations are shown in Figure 14.21.

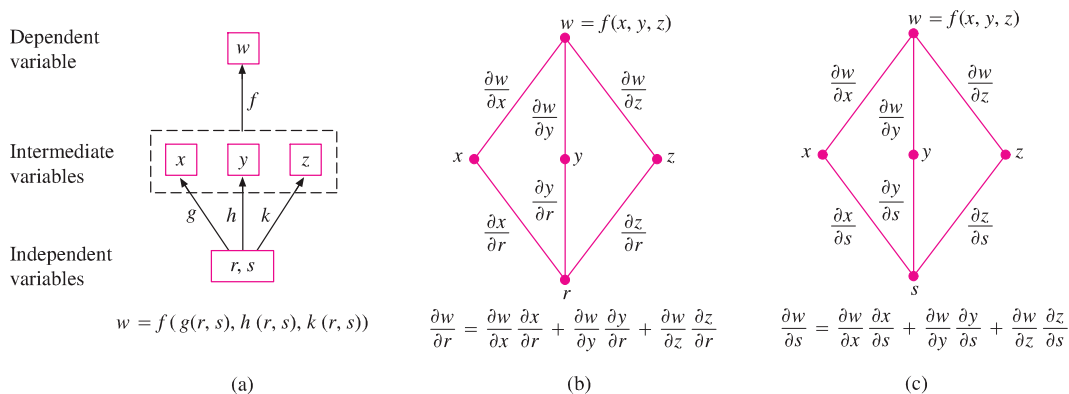


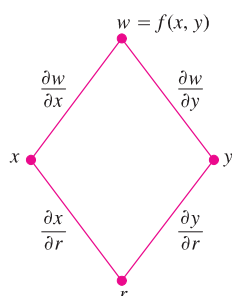
FIGURE 14.21 Composite function and branch diagrams for Theorem 7.

EXAMPLE 3 Express $\partial w/\partial r$ and $\partial w/\partial s$ in terms of r and s if

$$w = x + 2y + z^2, \quad x = \frac{r}{s}, \quad y = r^2 + \ln s, \quad z = 2r.$$

Solution Using the formulas in Theorem 7, we find

$$\begin{aligned}\frac{\partial w}{\partial r} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial r} \\ &= (1) \left(\frac{1}{s} \right) + (2)(2r) + (2z)(2) \\ &= \frac{1}{s} + 4r + (4r)(2) = \frac{1}{s} + 12r \quad \text{Substitute for intermediate variable } z. \\ \frac{\partial w}{\partial s} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial s} + \frac{\partial w}{\partial z} \frac{\partial z}{\partial s} \\ &= (1) \left(-\frac{r}{s^2} \right) + (2) \left(\frac{1}{s} \right) + (2z)(0) = \frac{2}{s} - \frac{r}{s^2}\end{aligned}$$

Chain Rule

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r}$$

FIGURE 14.22 Branch diagram for the equation

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r}.$$

If f is a function of two variables instead of three, each equation in Theorem 7 becomes correspondingly one term shorter.

If $w = f(x, y)$, $x = g(r, s)$, and $y = h(r, s)$, then

$$\frac{\partial w}{\partial r} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r} \quad \text{and} \quad \frac{\partial w}{\partial s} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial s}.$$

Figure 14.22 shows the branch diagram for the first of these equations. The diagram for the second equation is similar; just replace r with s .

EXAMPLE 4 Express $\partial w/\partial r$ and $\partial w/\partial s$ in terms of r and s if

$$w = x^2 + y^2, \quad x = r - s, \quad y = r + s.$$

Solution The preceding discussion gives the following.

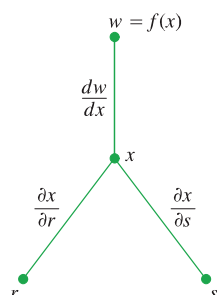
$$\begin{aligned} \frac{\partial w}{\partial r} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial r} & \frac{\partial w}{\partial s} &= \frac{\partial w}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial s} \\ &= (2x)(1) + (2y)(1) & &= (2x)(-1) + (2y)(1) \\ &= 2(r - s) + 2(r + s) & &= -2(r - s) + 2(r + s) \\ &= 4r & &= 4s \end{aligned}$$

Substitute for the intermediate variables.

If f is a function of x alone, our equations are even simpler.

If $w = f(x)$ and $x = g(r, s)$, then

$$\frac{\partial w}{\partial r} = \frac{dw}{dx} \frac{\partial x}{\partial r} \quad \text{and} \quad \frac{\partial w}{\partial s} = \frac{dw}{dx} \frac{\partial x}{\partial s}.$$

Chain Rule

$$\frac{\partial w}{\partial r} = \frac{dw}{dx} \frac{\partial x}{\partial r}$$

$$\frac{\partial w}{\partial s} = \frac{dw}{dx} \frac{\partial x}{\partial s}$$

FIGURE 14.23 Branch diagram for differentiating f as a composite function of r and s with one intermediate variable.

In this case, we use the ordinary (single-variable) derivative, dw/dx . The branch diagram is shown in Figure 14.23.

Implicit Differentiation Revisited

The two-variable Chain Rule in Theorem 5 leads to a formula that takes some of the algebra out of implicit differentiation. Suppose that

1. The function $F(x, y)$ is differentiable and
2. The equation $F(x, y) = 0$ defines y implicitly as a differentiable function of x , say $y = h(x)$.

Since $w = F(x, y) = 0$, the derivative dw/dx must be zero. Computing the derivative from the Chain Rule (branch diagram in Figure 14.24), we find

$$\begin{aligned} 0 &= \frac{dw}{dx} = F_x \frac{dx}{dx} + F_y \frac{dy}{dx} && \text{Theorem 5 with } t = x \text{ and } f = F \\ &= F_x \cdot 1 + F_y \cdot \frac{dy}{dx}. \end{aligned}$$

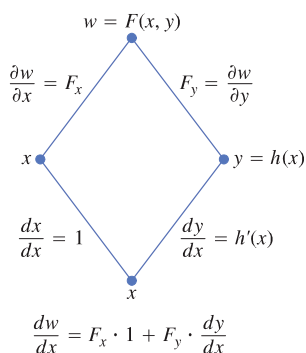


FIGURE 14.24 Branch diagram for differentiating $w = F(x, y)$ with respect to x . Setting $dw/dx = 0$ leads to a simple computational formula for implicit differentiation (Theorem 8).

If $F_y = \partial w / \partial y \neq 0$, we can solve this equation for dy/dx to get

$$\frac{dy}{dx} = -\frac{F_x}{F_y}.$$

We state this result formally.

THEOREM 8—A Formula for Implicit Differentiation Suppose that $F(x, y)$ is differentiable and that the equation $F(x, y) = 0$ defines y as a differentiable function of x . Then at any point where $F_y \neq 0$,

$$\frac{dy}{dx} = -\frac{F_x}{F_y}. \quad (1)$$

EXAMPLE 5 Use Theorem 8 to find dy/dx if $y^2 - x^2 - \sin xy = 0$.

Solution Take $F(x, y) = y^2 - x^2 - \sin xy$. Then

$$\begin{aligned} \frac{dy}{dx} &= -\frac{F_x}{F_y} = -\frac{-2x - y \cos xy}{2y - x \cos xy} \\ &= \frac{2x + y \cos xy}{2y - x \cos xy}. \end{aligned}$$

This calculation is significantly shorter than a single-variable calculation using implicit differentiation. ■

The result in Theorem 8 is easily extended to three variables. Suppose that the equation $F(x, y, z) = 0$ defines the variable z implicitly as a function $z = f(x, y)$. Then for all (x, y) in the domain of f , we have $F(x, y, f(x, y)) = 0$. Assuming that F and f are differentiable functions, we can use the Chain Rule to differentiate the equation $F(x, y, z) = 0$ with respect to the independent variable x :

$$\begin{aligned} 0 &= \frac{\partial F}{\partial x} \frac{\partial x}{\partial x} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial x} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial x} \\ &= F_x \cdot 1 + F_y \cdot 0 + F_z \cdot \frac{\partial z}{\partial x}, \end{aligned}$$

y is constant when differentiating with respect to x .

so

$$F_x + F_z \frac{\partial z}{\partial x} = 0.$$

A similar calculation for differentiating with respect to the independent variable y gives

$$F_y + F_z \frac{\partial z}{\partial y} = 0.$$

Whenever $F_z \neq 0$, we can solve these last two equations for the partial derivatives of $z = f(x, y)$ to obtain

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z}. \quad (2)$$

An important result from advanced calculus, called the **Implicit Function Theorem**, states the conditions for which our results in Equations (2) are valid. If the partial derivatives F_x , F_y , and F_z are continuous throughout an open region R in space containing the point (x_0, y_0, z_0) , and if for some constant c , $F(x_0, y_0, z_0) = c$ and $F_z(x_0, y_0, z_0) \neq 0$, then the equation $F(x, y, z) = c$ defines z implicitly as a differentiable function of x and y near (x_0, y_0, z_0) , and the partial derivatives of z are given by Equations (2).

EXAMPLE 6 Find $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$ at $(0, 0, 0)$ if $x^3 + z^2 + ye^{xz} + z \cos y = 0$.

Solution Let $F(x, y, z) = x^3 + z^2 + ye^{xz} + z \cos y$. Then

$$F_x = 3x^2 + zye^{xz}, \quad F_y = e^{xz} - z \sin y, \quad \text{and} \quad F_z = 2z + xye^{xz} + \cos y.$$

Since $F(0, 0, 0) = 0$, $F_z(0, 0, 0) = 1 \neq 0$, and all first partial derivatives are continuous, the Implicit Function Theorem says that $F(x, y, z) = 0$ defines z as a differentiable function of x and y near the point $(0, 0, 0)$. From Equations (2),

$$\frac{\partial z}{\partial x} = -\frac{F_x}{F_z} = -\frac{3x^2 + zye^{xz}}{2z + xye^{xz} + \cos y} \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{F_y}{F_z} = -\frac{e^{xz} - z \sin y}{2z + xye^{xz} + \cos y}.$$

At $(0, 0, 0)$ we find

$$\frac{\partial z}{\partial x} = -\frac{0}{1} = 0 \quad \text{and} \quad \frac{\partial z}{\partial y} = -\frac{1}{1} = -1. \quad \blacksquare$$

Functions of Many Variables

We have seen several different forms of the Chain Rule in this section, but each one is just a special case of one general formula. When solving particular problems, it may help to draw the appropriate branch diagram by placing the dependent variable on top, the intermediate variables in the middle, and the selected independent variable at the bottom. To find the derivative of the dependent variable with respect to the selected independent variable, start at the dependent variable and read down each route of the branch diagram to the independent variable, calculating and multiplying the derivatives along each route. Then add the products found for the different routes.

In general, suppose that $w = f(x, y, \dots, v)$ is a differentiable function of the variables $x, y, \dots, v$ (a finite set) and the $x, y, \dots, v$ are differentiable functions of $p, q, \dots, t$ (another finite set). Then w is a differentiable function of the variables p through t , and the partial derivatives of w with respect to these variables are given by equations of the form

$$\frac{\partial w}{\partial p} = \frac{\partial w}{\partial x} \frac{\partial x}{\partial p} + \frac{\partial w}{\partial y} \frac{\partial y}{\partial p} + \cdots + \frac{\partial w}{\partial v} \frac{\partial v}{\partial p}.$$

The other equations are obtained by replacing p by $q, \dots, t$, one at a time.

One way to remember this equation is to think of the right-hand side as the dot product of two vectors with components

$$\underbrace{\left(\frac{\partial w}{\partial x}, \frac{\partial w}{\partial y}, \dots, \frac{\partial w}{\partial v} \right)}_{\text{Derivatives of } w \text{ with respect to the intermediate variables}} \quad \text{and} \quad \underbrace{\left(\frac{\partial x}{\partial p}, \frac{\partial y}{\partial p}, \dots, \frac{\partial v}{\partial p} \right)}_{\text{Derivatives of the intermediate variables with respect to the selected independent variable}}.$$

Exercises 14.4

Chain Rule: One Independent Variable

In Exercises 1–6, (a) express dw/dt as a function of t , both by using the Chain Rule and by expressing w in terms of t and differentiating directly with respect to t . Then (b) evaluate dw/dt at the given value of t .

- $w = x^2 + y^2$, $x = \cos t$, $y = \sin t$; $t = \pi$
- $w = x^2 + y^2$, $x = \cos t + \sin t$, $y = \cos t - \sin t$; $t = 0$
- $w = \frac{x}{z} + \frac{y}{z}$, $x = \cos^2 t$, $y = \sin^2 t$, $z = 1/t$; $t = 3$
- $w = \ln(x^2 + y^2 + z^2)$, $x = \cos t$, $y = \sin t$, $z = 4\sqrt{t}$; $t = 3$
- $w = 2ye^x - \ln z$, $x = \ln(t^2 + 1)$, $y = \tan^{-1} t$, $z = e^t$; $t = 1$
- $w = z - \sin xy$, $x = t$, $y = \ln t$, $z = e^{t-1}$; $t = 1$

Chain Rule: Two and Three Independent Variables

In Exercises 7 and 8, (a) express $\partial z/\partial u$ and $\partial z/\partial v$ as functions of u and v both by using the Chain Rule and by expressing z directly in terms of u and v before differentiating. Then (b) evaluate $\partial z/\partial u$ and $\partial z/\partial v$ at the given point (u, v) .

- $z = 4e^x \ln y$, $x = \ln(u \cos v)$, $y = u \sin v$;
(a) $(u, v) = (2, \pi/4)$
- $z = \tan^{-1}(x/y)$, $x = u \cos v$, $y = u \sin v$;
(a) $(u, v) = (1.3, \pi/6)$

In Exercises 9 and 10, (a) express $\partial w/\partial u$ and $\partial w/\partial v$ as functions of u and v both by using the Chain Rule and by expressing w directly in terms of u and v before differentiating. Then (b) evaluate $\partial w/\partial u$ and $\partial w/\partial v$ at the given point (u, v) .

- $w = xy + yz + xz$, $x = u + v$, $y = u - v$, $z = uv$;
(a) $(u, v) = (1/2, 1)$
- $w = \ln(x^2 + y^2 + z^2)$, $x = ue^v \sin u$, $y = ue^v \cos u$,
 $z = ue^v$; (a) $(u, v) = (-2, 0)$

In Exercises 11 and 12, (a) express $\partial u/\partial x$, $\partial u/\partial y$, and $\partial u/\partial z$ as functions of x , y , and z both by using the Chain Rule and by expressing u directly in terms of x , y , and z before differentiating. Then (b) evaluate $\partial u/\partial x$, $\partial u/\partial y$, and $\partial u/\partial z$ at the given point (x, y, z) .

- $u = \frac{p-q}{q-r}$, $p = x + y + z$, $q = x - y + z$,
 $r = x + y - z$; (a) $(x, y, z) = (\sqrt{3}, 2, 1)$
- $u = e^{qr} \sin^{-1} p$, $p = \sin x$, $q = z^2 \ln y$, $r = 1/z$;
(a) $(x, y, z) = (\pi/4, 1/2, -1/2)$

Using a Branch Diagram

In Exercises 13–24, draw a branch diagram and write a Chain Rule formula for each derivative.

- $\frac{dz}{dt}$ for $z = f(x, y)$, $x = g(t)$, $y = h(t)$
- $\frac{dz}{dt}$ for $z = f(u, v, w)$, $u = g(t)$, $v = h(t)$, $w = k(t)$
- $\frac{\partial w}{\partial u}$ and $\frac{\partial w}{\partial v}$ for $w = h(x, y, z)$, $x = f(u, v)$, $y = g(u, v)$,
 $z = k(u, v)$

- $\frac{\partial w}{\partial x}$ and $\frac{\partial w}{\partial y}$ for $w = f(r, s, t)$, $r = g(x, y)$, $s = h(x, y)$,
 $t = k(x, y)$
- $\frac{\partial w}{\partial u}$ and $\frac{\partial w}{\partial v}$ for $w = g(x, y)$, $x = h(u, v)$, $y = k(u, v)$
- $\frac{\partial w}{\partial x}$ and $\frac{\partial w}{\partial y}$ for $w = g(u, v)$, $u = h(x, y)$, $v = k(x, y)$
- $\frac{\partial z}{\partial t}$ and $\frac{\partial z}{\partial s}$ for $z = f(x, y)$, $x = g(t, s)$, $y = h(t, s)$
- $\frac{\partial y}{\partial r}$ for $y = f(u)$, $u = g(r, s)$
- $\frac{\partial w}{\partial s}$ and $\frac{\partial w}{\partial t}$ for $w = g(u)$, $u = h(s, t)$
- $\frac{\partial w}{\partial p}$ for $w = f(x, y, z, v)$, $x = g(p, q)$, $y = h(p, q)$,
 $z = j(p, q)$, $v = k(p, q)$
- $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial s}$ for $w = f(x, y)$, $x = g(r)$, $y = h(s)$
- $\frac{\partial w}{\partial s}$ for $w = g(x, y)$, $x = h(r, s, t)$, $y = k(r, s, t)$

Implicit Differentiation

Assuming that the equations in Exercises 25–28 define y as a differentiable function of x , use Theorem 8 to find the value of dy/dx at the given point.

- $x^3 - 2y^2 + xy = 0$, $(1, 1)$
- $xy + y^2 - 3x - 3 = 0$, $(-1, 1)$
- $x^2 + xy + y^2 - 7 = 0$, $(1, 2)$
- $xe^y + \sin xy + y - \ln 2 = 0$, $(0, \ln 2)$

Find the values of $\partial z/\partial x$ and $\partial z/\partial y$ at the points in Exercises 29–32.

- $z^3 - xy + yz + y^3 - 2 = 0$, $(1, 1, 1)$
- $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} - 1 = 0$, $(2, 3, 6)$
- $\sin(x + y) + \sin(y + z) + \sin(x + z) = 0$, (π, π, π)
- $xe^y + ye^z + 2 \ln x - 2 - 3 \ln 2 = 0$, $(1, \ln 2, \ln 3)$

Finding Partial Derivatives at Specified Points

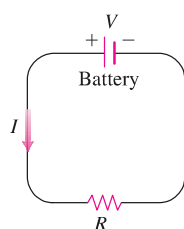
- Find $\partial w/\partial r$ when $r = 1, s = -1$ if $w = (x + y + z)^2$,
 $x = r - s, y = \cos(r + s), z = \sin(r + s)$.
- Find $\partial w/\partial v$ when $u = -1, v = 2$ if $w = xy + \ln z$,
 $x = v^2/u, y = u + v, z = \cos u$.
- Find $\partial w/\partial v$ when $u = 0, v = 0$ if $w = x^2 + (y/x)$,
 $x = u - 2v + 1, y = 2u + v - 2$.
- Find $\partial z/\partial u$ when $u = 0, v = 1$ if $z = \sin xy + x \sin y$,
 $x = u^2 + v^2, y = uv$.
- Find $\partial z/\partial u$ and $\partial z/\partial v$ when $u = \ln 2, v = 1$ if $z = 5 \tan^{-1} x$ and $x = e^u + \ln v$.
- Find $\partial z/\partial u$ and $\partial z/\partial v$ when $u = 1, v = -2$ if $z = \ln q$ and $q = \sqrt{v + 3 \tan^{-1} u}$.

Theory and Examples

39. Assume that $w = f(s^3 + t^2)$ and $f'(x) = e^x$. Find $\frac{\partial w}{\partial t}$ and $\frac{\partial w}{\partial s}$.
40. Assume that $w = f\left(ts^2, \frac{s}{t}\right)$, $\frac{\partial f}{\partial x}(x, y) = xy$, and $\frac{\partial f}{\partial y}(x, y) = \frac{x^2}{2}$. Find $\frac{\partial w}{\partial t}$ and $\frac{\partial w}{\partial s}$.
41. **Changing voltage in a circuit** The voltage V in a circuit that satisfies the law $V = IR$ is slowly dropping as the battery wears out. At the same time, the resistance R is increasing as the resistor heats up. Use the equation

$$\frac{dV}{dt} = \frac{\partial V}{\partial I} \frac{dI}{dt} + \frac{\partial V}{\partial R} \frac{dR}{dt}$$

to find how the current is changing at the instant when $R = 600$ ohms, $I = 0.04$ amp, $dR/dt = 0.5$ ohm/sec, and $dV/dt = -0.01$ volt/sec.



42. **Changing dimensions in a box** The lengths a , b , and c of the edges of a rectangular box are changing with time. At the instant in question, $a = 1$ m, $b = 2$ m, $c = 3$ m, $da/dt = db/dt = 1$ m/sec, and $dc/dt = -3$ m/sec. At what rates are the box's volume V and surface area S changing at that instant? Are the box's interior diagonals increasing in length or decreasing?
43. If $f(u, v, w)$ is differentiable and $u = x - y$, $v = y - z$, and $w = z - x$, show that

$$\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} = 0.$$

44. **Polar coordinates** Suppose that we substitute polar coordinates $x = r \cos \theta$ and $y = r \sin \theta$ in a differentiable function $w = f(x, y)$.

a. Show that

$$\frac{\partial w}{\partial r} = f_x \cos \theta + f_y \sin \theta$$

and

$$\frac{1}{r} \frac{\partial w}{\partial \theta} = -f_x \sin \theta + f_y \cos \theta.$$

- b. Solve the equations in part (a) to express f_x and f_y in terms of $\partial w / \partial r$ and $\partial w / \partial \theta$.
- c. Show that

$$(f_x)^2 + (f_y)^2 = \left(\frac{\partial w}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial w}{\partial \theta}\right)^2.$$

45. **Laplace equations** Show that if $w = f(u, v)$ satisfies the Laplace equation $f_{uu} + f_{vv} = 0$ and if $u = (x^2 - y^2)/2$ and $v = xy$, then w satisfies the Laplace equation $w_{xx} + w_{yy} = 0$.
46. **Laplace equations** Let $w = f(u) + g(v)$, where $u = x + iy$, $v = x - iy$, and $i = \sqrt{-1}$. Show that w satisfies the Laplace equation $w_{xx} + w_{yy} = 0$ if all the necessary functions are differentiable.

47. **Extreme values on a helix** Suppose that the partial derivatives of a function $f(x, y, z)$ at points on the helix $x = \cos t$, $y = \sin t$, $z = t$ are

$$f_x = \cos t, \quad f_y = \sin t, \quad f_z = t^2 + t - 2.$$

At what points on the curve, if any, can f take on extreme values?

48. **A space curve** Let $w = x^2 e^{2y} \cos 3z$. Find the value of dw/dt at the point $(1, \ln 2, 0)$ on the curve $x = \cos t$, $y = \ln(t + 2)$, $z = t$.

49. **Temperature on a circle** Let $T = f(x, y)$ be the temperature at the point (x, y) on the circle $x = \cos t$, $y = \sin t$, $0 \leq t \leq 2\pi$ and suppose that

$$\frac{\partial T}{\partial x} = 8x - 4y, \quad \frac{\partial T}{\partial y} = 8y - 4x.$$

- a. Find where the maximum and minimum temperatures on the circle occur by examining the derivatives dT/dt and d^2T/dt^2 .
- b. Suppose that $T = 4x^2 - 4xy + 4y^2$. Find the maximum and minimum values of T on the circle.

50. **Temperature on an ellipse** Let $T = g(x, y)$ be the temperature at the point (x, y) on the ellipse

$$x = 2\sqrt{2} \cos t, \quad y = \sqrt{2} \sin t, \quad 0 \leq t \leq 2\pi,$$

and suppose that

$$\frac{\partial T}{\partial x} = y, \quad \frac{\partial T}{\partial y} = x.$$

- a. Locate the maximum and minimum temperatures on the ellipse by examining dT/dt and d^2T/dt^2 .
- b. Suppose that $T = xy - 2$. Find the maximum and minimum values of T on the ellipse.

Differentiating Integrals Under mild continuity restrictions, it is true that if

$$F(x) = \int_a^b g(t, x) dt,$$

then $F'(x) = \int_a^b g_x(t, x) dt$. Using this fact and the Chain Rule, we can find the derivative of

$$F(x) = \int_a^{f(x)} g(t, x) dt$$

by letting

$$G(u, x) = \int_a^u g(t, x) dt,$$

where $u = f(x)$. Find the derivatives of the functions in Exercises 51 and 52.

$$51. F(x) = \int_0^{x^2} \sqrt{t^4 + x^3} dt \quad 52. F(x) = \int_{x^2}^1 \sqrt{t^3 + x^2} dt$$